

Design of Silicon Nitride Polarization Beam Splitter using a Directional Coupler

Fangwei Chang*

*Department of Electrical and Computer Engineering, University of Toronto

Abstract—This project demonstrates the design of project demonstrates the design of a silicon nitride (Si_3N_4) polarization beam splitter at 1520 nm, using a directional coupler (DC) design. The device is designed for single mode operation, separating a TEM input input into a TE and TM mode output. Calculations are performed using online waveguide solvers. Possible fabrication steps and assessments of losses are also discussed.

Index Terms—Waveguide, silicon nitride, silicon on insulator, polarization splitter, direct coupler, integrated optical circuit.

I. INTRODUCTION

THE polarization beam splitter separates different polarizations of electromagnetic waves into distinct outputs. Such devices are useful for a variety of applications including communications, imaging and signal processing [1] [2]. Silicon nitride offers lower losses (compared to silicon), low temperature sensitivity and compatibility with complimentary metal-oxide semiconductor (CMOS) structures, making it a desirable waveguide material.

The basic topology of a directional coupler (DC) polarization beam splitter is shown in fig. 1. The input consists of TEM polarized light in a single waveguide. By placing an adjacent waveguide to the input waveguide, the two couple producing even (ESM) and odd supermodes (ESM, OSM respectively) which propagate at different velocities. After travelling some coupling length L_c , the light is transferred completely to the other waveguide when the supermodes are π out of phase (fig. 2, 3) [5]. By adjusting L_c , one polarization can be separated from the other. The two output waveguides are then bent away from the other until the two are separated by the desired distance.

In this project, the waveguides are designed for single-mode operation at 1520 nm, i.e. designed to support the TE₀ and TM₀ modes. The output consists of two waveguides separated by 50 μm . A silicon (Si) substrate is used for mechanical support, with a silicon dioxide (SiO_2) insulation layer; the silicon nitride (Si_3N_4) forming the waveguides is deposited and etched on top. A coating of SiO_2 is then applied on top, completely surrounding the Si_3N_4 as shown in fig. 1. At the design wavelength of 1520 nm, the index of SiO_2 is ~ 1.444 and the index of silicon nitride is ~ 1.997 [6].

A. General Design Procedure

Online solvers publicly available at the "Simulations in Integrated Optics" (SiIo) website are used to perform the design [7]. The approximate steps are as follows:

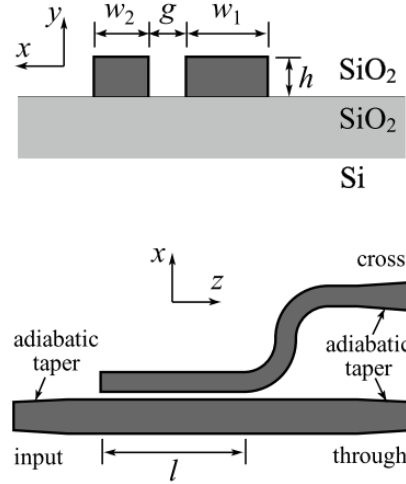


Fig. 1. Example of a basic polarization splitter topology, adopted in this project, showing waveguide (dark grey) seated on an SiO_2 isolation layer with a SiO_2 coating on top. The Si bottom layer acts as structural support. The input arrives and is split into two output waveguides. l is the coupling length, g is the waveguide separation, h is the waveguide height. Figure modified from [4].

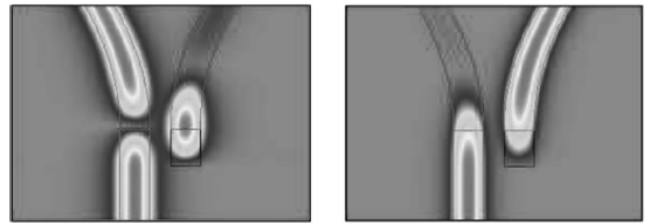


Fig. 2. Example of a simulated polarization splitter in operation, showing the separation of the TE (left) and TM modes (right) as they couple into the different output waveguides. Here $L_{c,\text{TM}}$ is designed to be 2 times that of $L_{c,\text{TE}}$ [1].

- 1) "OMS" 1D waveguide solver is used to determine the maximum h for a slab waveguide satisfying the single mode condition.
- 2) "EMIS" 2D waveguide solver is used to determine the waveguide widths w_1 , w_2 and separation g (fig. 1) to achieve the polarization separation effect, using h from the previous part.
- 3) "BendS" solver for curved dielectric interfaces is used to estimate bend losses.

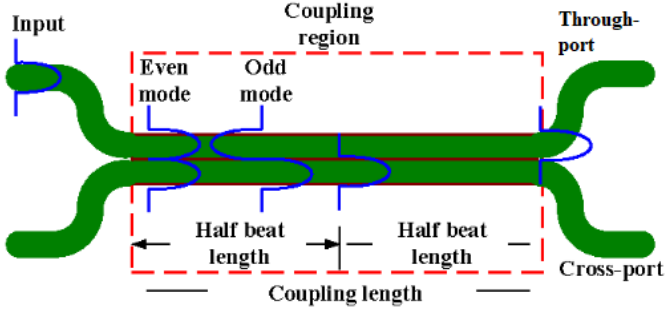


Fig. 3. Diagram of even supermode (ESM) and odd supermode (OSM) excitation between two adjacent waveguides. Figure modified from [3].

TABLE I
DIMENSIONS (μm) FOR SYMMETRIC COUPLED WAVEGUIDES, $\lambda = 1520$ nm (SEE FIG. 1).

h , height	0.30
w , width	0.69
g , separation	1.0

Trial-and-error refinement of parameters is used when necessary to determine the final design. Details and motivations of the design parameters are discussed in the subsequent sections.

II. WAVEGUIDE DESIGN

The input waveguide is designed to propagate only the TE₀/TM₀ mode. To satisfy the single-mode condition, the normalized frequency number V must fulfill:

$$V < \pi \quad (1)$$

$$\Rightarrow \frac{2\pi h}{\lambda} \sqrt{n_1^2 - n_2^2} < \pi \quad (2)$$

Where h is the core thickness (height of the waveguide), n_1 is the index of the waveguide material (Si_3N_4 , $n \simeq 1.997$) and n_2 is the index of the cladding (SiO_2 , $n \simeq 1.444$). For $\lambda = 1520$ nm, $h \leq 0.55 \mu\text{m}$. The result is verified using the OMS solver (fig. 4); as expected, $h = 0.55 \mu\text{m}$ supports only the TE₀ and TM₀ modes, whereas any greater h leads to undesired TE₁ and TM₁ modes arising.

III. WAVEGUIDE COUPLER DESIGN

The coupling length L_c is the waveguide length necessary for total energy transfer, which is different from TE and TM polarization. In general, L_c is given by:

$$L_c = \frac{\pi}{\beta_e - \beta_o} = \frac{\pi}{k_o(n_e - n_o)} \quad (3)$$

Where k_o is the free-space wavelength, and n_e and n_o are the refractive indices of the even and odd supermodes (β is the phase constant); this calculation is implemented in the EMIS solver. The L_c for one of the TE/TM polarizations should be twice as long as the other. Thus, for an integer m [1]:

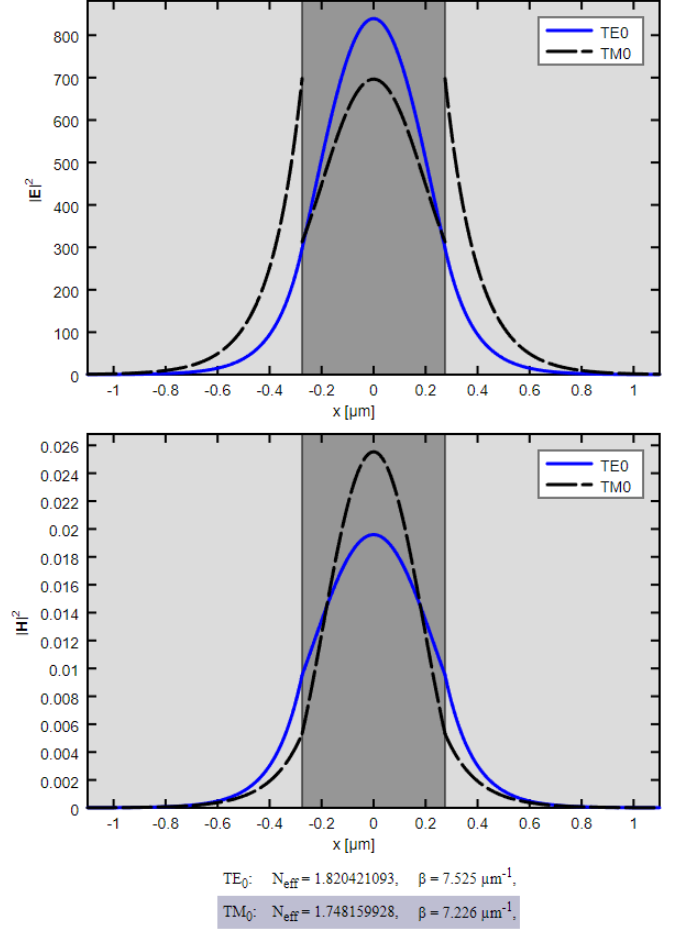


Fig. 4. Field amplitudes for Si_3N_4 (dark grey, cladding in light grey) slab waveguide with core thickness of $h = 0.55 \mu\text{m}$ for single-mode operation. Effective indices of the TE₀, TM₀ modes as well as propagation constants are shown. Computed using OMS solver [7].

$$\frac{L_{c,\text{TM}}}{L_{c,\text{TE}}} = 2m \quad \text{or} \quad (4)$$

$$\frac{L_{c,\text{TE}}}{L_{c,\text{TM}}} = 2m \quad (5)$$

To minimize propagation losses, use of material and device size, $m = 1$ is ideal [4]. The polarization with the shorter L_c fully transfers to the "cross" waveguide and then back to the "through" waveguide, whereas for the polarization with the longer L_c there is only sufficient distance to transfer to the "cross" waveguide and not back again (fig. 1, 3). Thus they are separated. L_c depends exponentially on the gap separation g (see fig. 1) of the device [5]; the parameters are manually fine-tuned to achieve the final value.

A pair of identical waveguides are used for the coupling section and designed using the EMIS solver [7], resulting in $L_{c,\text{TE}} = 111 \mu\text{m}$ (coupling length between TE₀₀, TE₀₁ modes) and $L_{c,\text{TM}} = 54.1 \mu\text{m}$ (coupling length between TM₀₀, TM₀₁ modes). Thus $L_{c,\text{TE}}$ is the length of the coupling section in the device; the TE mode is transferred to the cross

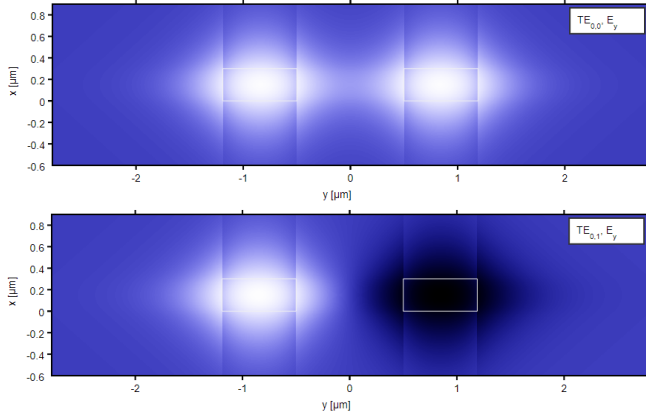


Fig. 5. Demonstration of TE00 and TE01 supermode for coupling section. Waveguide dimensions and separation are listed in table I. Computed using EMIS solver [7].

$L_c/\mu\text{m}$	TE _{0,0}	TE _{0,1}	TM _{0,0}	TM _{0,1}
TE _{0,0}	×	111	28.8	18.8
TE _{0,1}	111	×	38.8	22.6
TM _{0,0}	28.8	38.8	×	54.1
TM _{0,1}	18.8	22.6	54.1	×

TE _{0,0} :	$N_{\text{eff}} = 1.511538559,$	$\beta = 6.248208464 \mu\text{m}^{-1}$
TE _{0,1} :	$N_{\text{eff}} = 1.50470515,$	$\beta = 6.219961374 \mu\text{m}^{-1}$
TM _{0,0} :	$N_{\text{eff}} = 1.485133201,$	$\beta = 6.13905731 \mu\text{m}^{-1}$
TM _{0,1} :	$N_{\text{eff}} = 1.47109256,$	$\beta = 6.081017868 \mu\text{m}^{-1}$

Fig. 6. Coupling lengths between supermodes, effective indices and propagation constants. $L_{c,\text{TE}}$ for TE00 $\times$ TE01 is approximately twice that of $L_{c,\text{TM}}$ for TM00 $\times$ TM01 as desired. No unwanted high modes are supported by the waveguides. Computed using EMIS solver [7].

waveguide output. The separation and waveguide dimensions are listed in table I.

IV. BEND DESIGN

After separation of the input polarizations are achieved, the output waveguides are directed away from each other by bending one or both. For simplicity, the topology in fig. 1 is adopted where one of the output waveguides ("cross" in fig. 1) is directed away by two $\frac{\pi}{2}$ bends. The radius of each bend is $r = 25 \mu\text{m}$ to achieve the required $50 \mu\text{m}$ separation, with the total length of each bend being $\frac{\pi r}{2} \simeq 39.26 \mu\text{m}$.

V. SOURCES OF LOSS

In this section, various sources of loss are discussed in order to assess the performance of the coupler design. Losses will be considered for the $111 \mu\text{m}$ section as well as the two $\frac{\pi}{2}$ degree bends. In many cases a rough estimation is possible by referring to sources in the literature, although using simulations is also possible.

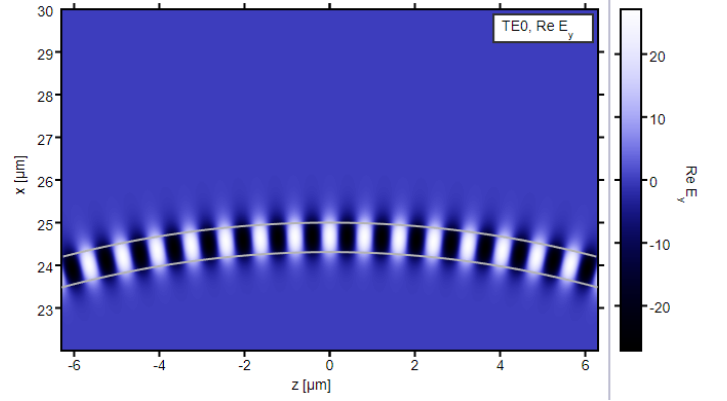


Fig. 7. $\text{Re}[E_y]$ field for TE0 mode in a segment of one $\frac{\pi}{2}$ bend. Computed using Bends solver [7].

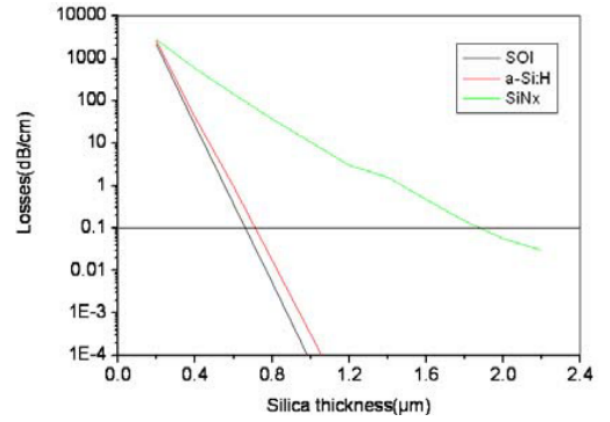


Fig. 8. Substrate leakage loss for various thicknesses of SiO_2 over Si substrate, for different waveguide materials [9].

The estimated additional loss due to the polarization splitter for the TE0 mode is 0.17 dB (reduction to 0.961 of initial amplitude). For the TM0 mode, it is 0.01 dB (reduction to 0.997 of initial amplitude).

A. Intrinsic Propagation Loss

Propagation loss is due to the intrinsic properties of the Si_3N_4 is considered. For Si_3N_4 waveguides, attenuation of 1.2 dB/cm at 1520 nm is reported (reduction to ~ 0.756 of initial amplitude), corresponding to approximately 0.01 dB loss for the $111 \mu\text{m}$ coupling section (reduction to ~ 0.997 of initial amplitude) [2] [8].

B. Substrate Leakage Loss

Substrate leakage loss to the Si substrate can be calculated using coupled-mode theory (CMT) or numeric simulation. Substrate leakage loss can be reduced by increasing the thickness of the SiO_2 isolation layer between the substrate and the waveguide. For Si_3N_4 at approximately $\lambda = 1.5 \mu\text{m}$, a $2 \mu\text{m}$ SiO_2 layer leads to < 0.1 dB/cm substrate leakage loss, or 0.001 dB for $L_c = 111 \mu\text{m}$ (reduction to ~ 0.9997 of initial amplitude, fig. 8) [9].

C. Bend Loss

Introducing bends in the waveguide causes loss, which increases with poorer confinement caused by shorter bend radius and low index contrast. In the proposed design, the TE0 mode is outputted through the "cross" waveguide and is affected by the bend. Using the BendS online solver, the TE0 propagation constant in one $\frac{\pi}{2}$ bend is approximately $\gamma = (7.60 - i1.59 \times 10^{-15}) \mu\text{m}^{-1}$.

At approximately $\lambda = 1.5 \mu\text{m}$, bend losses for a $\frac{\pi}{2}$ bend with $r = 20 \mu\text{m}$ is 0.06 dB (reduction to ~ 0.986279 of initial amplitude) [10]. Interpolating, the estimated bend loss for $r = 25 \mu\text{m}$ is, with two consecutive bends producing a total loss of 0.15 dB (reduction to ~ 0.966 of initial amplitude).

VI. PROPOSED FABRICATION

Fabrication of the polarization splitter can be performed using conventional optical lithography [11]. The proposed fabrication steps are as follows (partially illustrated in fig. 9):

- 1) Si substrate is cleaned using hydrochloric acid (HCl) to remove oxidation. Ultrasound cleaning is performed in acetone, methanol and isopropanol for 5 minutes each. The wafer is air dried with nitrogen gas, then baked at 120 °C for 5 minutes.
- 2) Plasma-enhanced metalorganic chemical vapor deposition (MOCVD) is used to apply a 2 μm SiO₂ isolation layer to the Si substrate.
- 3) MOCVD is used to apply a 0.3 μm Si₃N₄ layer over the SiO₂ isolation layer.
- 4) Wafer is spin-coated with 1 μm photoresist (e.x. SU-8) at 1500-8000 rpm, then baked at 75 - 100 °C.
- 5) Laser-fabricated photomask is aligned with the wafer. Optical lithography is used to imprint the desired waveguide pattern onto the photoresist, using ultraviolet (UV) light at 365 nm.
- 6) Dry chemical etching (DCE) is used to produce the desired waveguide pattern. A solvent (e.x. acetone) is used to remove the remaining photoresist layer.
- 7) Developing solution (e.x. n-amyl acetate) is used to remove UV exposed photoresist. Additional baking removes leftover solvent residues.
- 8) MOCVD is used to apply a 1 μm SiO₂ cladding layer over the waveguide, completing the fabrication.

Precise fabrication ensures optimal performance and minimizes scattering loss from rough surfaces. The critical dimension (CD) is the smallest physical feature in the device and determines the resolution required to fabricate it; this is the 1 μm gap g in the case of this design. The minimum resolution of optical lithography is down to 100 nm, which is not a concern for the splitter design in this project. Sensitivity to fabrication error should be within normal limits, as no non-standard techniques are used.

VII. CONCLUSION

In this project, the design of a simple Si₃N₄ DC polarization splitter for single-mode operation at 1520 nm is presented.

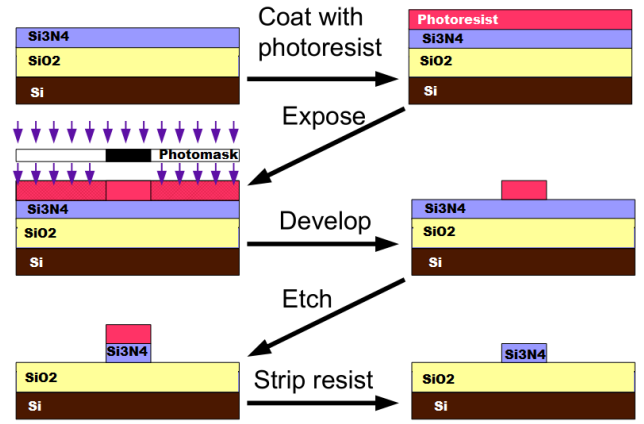


Fig. 9. Schematic steps 3-7 of the proposed fabrication procedure.

Losses are assessed and are within acceptable levels. Steps for possible fabrication are also discussed. Using Si₃N₄ offers several advantages over standard waveguides and is a desirable material for integrated optical circuits.

ACKNOWLEDGMENT

The author would like to thank Prof. Stewart Aitchison for his helpful teaching and guidance which allowed this project's completion.

REFERENCES

- [1] Kiyat, I., Aydinli, A., Nadir, D. A compact silicon-on-insulator MMI-based polarization splitter. 2007 IEEE/LEOS Int. Conf. Opt. MEMS Nanophotonics, OMENS 17, 121–122 (2007).
- [2] Blumenthal, D. J., Heideman, R., Geuzebroek, D., Leinse, A., Roeloffzen, C. Silicon Nitride in Silicon Photonics. Proc. IEEE 106, 2209–2231 (2018).
- [3] Yu, J. et al. High performance micro-fiber coupler-based polarizer and band-rejection filter. Opt. Express 20, 17258 (2012).
- [4] Yuxi, F. Polarization Beam Splitter Based on Si₃N₄/SiO₂ Horizontal Slot Waveguides for On-Chip High-Power Applications. Opt. Fiber Commun. Conf. 2016 20, 1–13 (2020).
- [5] Lee, S.-Y., Darmawan, S., Lee, C.-W., Chin, M.-K. Transformation between directional couplers and multi-mode interferometers based on ridge waveguides. Opt. Express 12, 3079 (2004).
- [6] M. Polyanskiy, Refractive Index Database. [Online]. Available: <https://refractiveindex.info/>. [Accessed: 21-Apr-2022].
- [7] M. Hammer, SiO: solver codes, manuals, online solvers. [Online]. Available: <https://www.sio.eu/index.html>. [Accessed: 22-Apr-2022].
- [8] Henry, C. H., Kazarinov, R. F. Low loss Si₃N₄-SiO₂ optical waveguides on Si. Appl. Opt. 26, 2621–2624 (1987).
- [9] Fan, G. et al. Optical Waveguides on Three Material Platforms of Silicon-on-Insulator, Amorphous Silicon and Silicon Nitride. IEEE J. Sel. Top. Quantum Electron. 22, 225–231 (2016).
- [10] Song, J. H. et al. Low-Loss Waveguide Bends by Advanced Shape for Photonic Integrated Circuits. J. Light. Technol. 38, 3273–3279 (2020).
- [11] Huang, Y. et al. Sub-micron silicon nitride waveguide fabrication using conventional optical lithography. Opt. InfoBase Conf. Pap. 1, 1–3 (2014).